Valvular Heart Disease — Surgery

The Big Picture

4 valves: 2 AV (mitral, tricuspid) + 2 semilunar (aortic, pulmonary — 3 cusps each). Disease = **stenosis** (won't open, pressure overload) or **regurgitation/insufficiency** (won't close, volume overload). Surgeon decides **repair vs replace**, and **which prosthetic valve**.

Aortic Regurgitation (AR)

- **What** — backflow into LV in diastole. From leaflet disease OR aortic root dilatation (cusps pulled apart).
- **Causes** — rheumatic fever, Marfan/annuloaortic ectasia, bicuspid, myxoid degeneration, infective endocarditis, trauma, aortic dissection.
- **Physiology** — volume overload; low diastolic pressure → ↓ coronary flow.
- **Symptoms** — dyspnoea, orthopnoea, PND, angina, CHF (often long asymptomatic).
- **Operate when** — symptomatic severe/acute; LV end-systolic >55mm; EF <50%; other cardiac surgery needed; significant LVH/dilatation.
- **Options** — repair, replacement (valve ± aortic conduit = "composite"), TAVI (mostly for AS, rarely AR).

Aortic Stenosis (AS)

- **What** — fused rigid stenotic cusps. Pressure overload on LV → LVH.
- **Causes — calcific/degenerative** (commonest), congenital (bicuspid), rheumatic.
- **Classic triad** — chest pain, dyspnoea, **syncope**.
- **Exam** — pulsus **parvus et tardus** (slow, weak); harsh systolic murmur at right 2nd ICS **radiating to carotids**; soft/absent S2 in severe.
- **Tests** — ECG (LVH), echo (severity), catheterisation if CAD risk before replacement.
- **Operate when** — symptomatic; or EF<50%, valve area <1cm², gradient >50mmHg, LVH.
- **NEVER give vasodilators in AS.** Balloon valvuloplasty = bridge/palliation only. **TAVI** is the catheter option.

Mitral Regurgitation (MR)

- **What** — LV→LA backflow in systole.
- **Causes** — myxomatous degeneration, annular calcification, rheumatic, ruptured chordae/papillary muscle, IE, Marfan, ischaemic, functional (dilated CM).
- **Physiology** — LVH, LV dilation, ↑LVEDP, heart failure.
- **Exam** — soft S1, **holosystolic blowing murmur at apex radiating to axilla**, widely split S2, displaced apex.
- **Management** — medical (anti-failure, digoxin, anticoagulant) then surgical: **repair**, ring annuloplasty, apparatus-sparing replacement, or replacement.
- **Operate** — symptomatic, severe structural damage, other surgery needed.

Mitral Stenosis (MS)

- **What** — valve won't open fully in diastole.
- **Cause — rheumatic (99%** of removed valves). Rare: degenerative, carcinoid, congenital (<1%).
- **Management** — medical + surgical: **balloon valvuloplasty**, repair (**commissurotomy**), apparatus-sparing replacement, replacement.

Choosing the Valve

- **Mechanical** — Ball-and-socket (Starr-Edwards), tilting disc, **bileaflet (St Jude)**. Durable but **lifelong anticoagulation**.
- **Tissue** — porcine, pericardial (bovine), homograft. No long-term anticoagulant but **wears out**.
- **Choice depends on** — age, **childbearing age**, anticoagulant contraindications, valve position, patient request/availability.

Walk-Away Points

- AS triad: **angina, syncope, dyspnoea**; murmur **radiates to carotids**; NO vasodilators.
- MR murmur **radiates to axilla**; MS is **rheumatic** (99%).
- Mechanical = durable + lifelong warfarin; tissue = no warfarin but degenerates → matters most in childbearing age.
- TAVI = mainly aortic stenosis.

Valvular Heart Disease — Revision Table

Lesion	Load	Top causes	Murmur / sign	Surgery indication
Aortic regurg (AR)	Volume	Rheumatic, Marfan, bicuspid, IE, dissection	Diastolic; wide pulse pressure	Sx severe/acute; ESD>55mm; EF<50%; LVH
Aortic stenosis (AS)	Pressure	**Calcific**, congenital bicuspid, rheumatic	Systolic at R 2nd ICS → **carotids**; pulsus parvus et tardus; soft S2	Sx; EF<50%; area<1cm²; gradient>50
Mitral regurg (MR)	Volume	Myxomatous, rheumatic, ruptured chordae, ischaemic	Holosystolic blowing at apex → **axilla**; soft S1	Sx; severe structural damage
Mitral stenosis (MS)	—	**Rheumatic 99%**	Diastolic rumble	Balloon/commissurotomy/replace

Op type	Use
Repair / annuloplasty / commissurotomy	Preserve native valve
Replacement (± conduit = composite)	Damaged valve ± aortic root
TAVI	Mainly aortic **stenosis**, catheter via femoral
Balloon valvuloplasty	Bridge/palliation (AS, MS)

Valve type	Pros	Cons
Mechanical (St Jude bileaflet, Starr-Edwards)	Durable, lifelong	**Lifelong anticoagulation**
Tissue (porcine, bovine pericardial, homograft)	No long anticoagulant	**Degenerates / wears out**

Valve choice depends on	
Age & **childbearing age**	Anticoagulant contraindications
Valve position	Patient request / availability

AS rules	
Classic triad	Angina, syncope, dyspnoea
NEVER	Vasodilators in AS

Bold = high-yield.